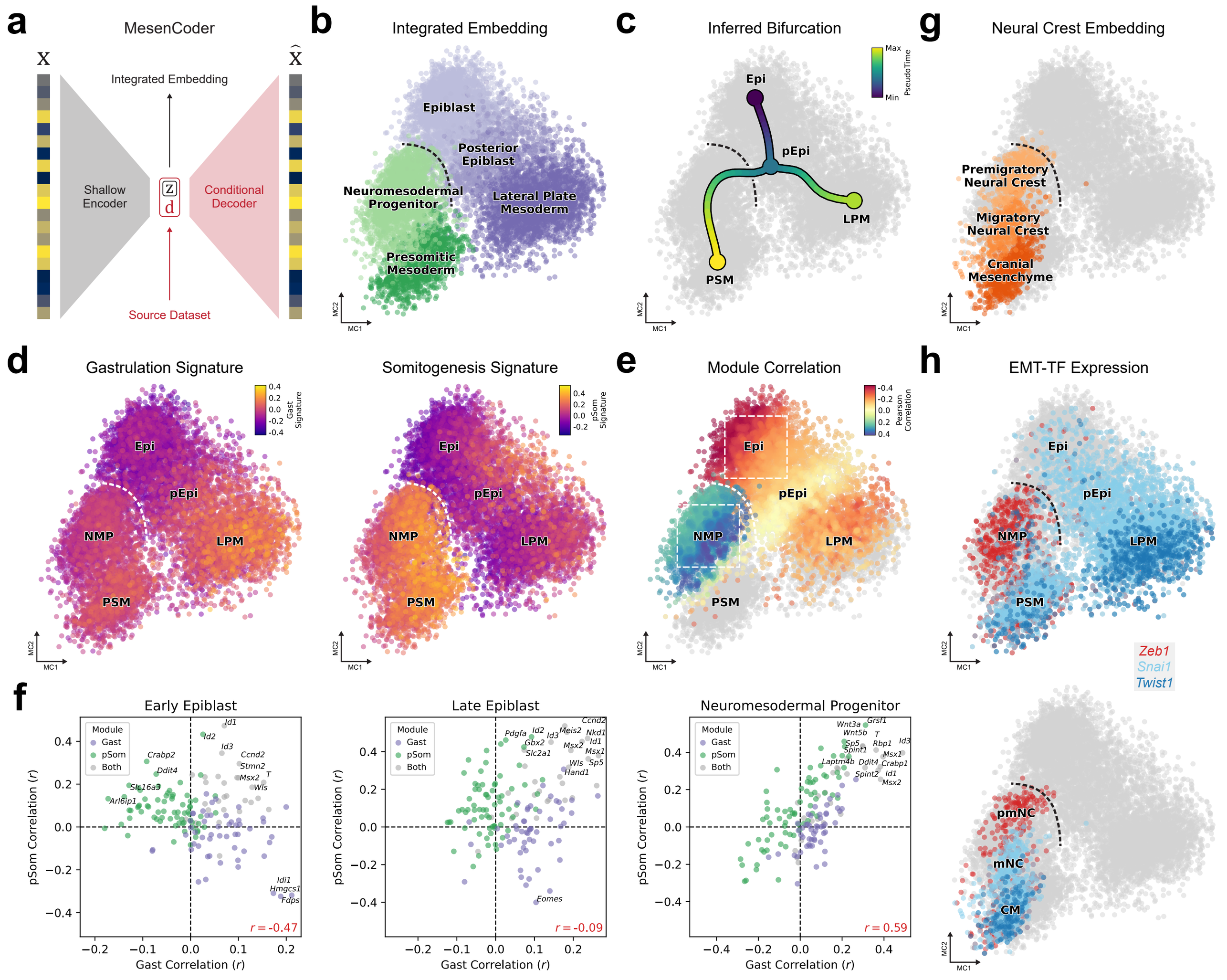

a
MesenCoder
x
x̂
Integrated Embedding
Shallow Encoder
z
d
Conditional Decoder
Source Dataset
b
Integrated Embedding
Epiblast
Posterior Epiblast
Neuromesodermal Progenitor
Lateral Plate Mesoderm
Presomitic Mesoderm
MC2
MC1
c
Inferred Bifurcation
Max
Min
PseudoTime
Epi
pEpi
LPM
PSM
MC2
MC1
g
Neural Crest Embedding
Premigratory Neural Crest
Migratory Neural Crest
Cranial Mesenchyme
MC2
MC1
d
Gastrulation Signature
0.4
0.2
0.0
-0.2
-0.4
Gast Signature
Epi
pEpi
NMP
LPM
PSM
MC2
MC1
Somitogenesis Signature
0.4
0.2
0.0
-0.2
pSom Signature
Epi
pEpi
NMP
LPM
PSM
MC2
MC1
e
Module Correlation
-0.4
-0.2
0.0
0.2
0.4
Pearson Correlation
Epi
pEpi
NMP
LPM
PSM
MC2
MC1
h
EMT-TF Expression
Epi
pEpi
NMP
LPM
PSM
MC2
MC1
Zeb1
Snai1
Twist1
f
Early Epiblast
Module
Gast
pSom
Both
Id1
Id2
Id3
Crabp2
Ccnd2
Ddit4
Stmn2
Msx2
T
Slc16a3
Wls
Arl6ip1
Idi1
Hmgcs1
Fdps
r = -0.47
pSom Correlation (r)
Gast Correlation (r)
Late Epiblast
Module
Gast
pSom
Both
Pdgfa
Id2
Id3
Meis2
Nkd1
Gbx2
Msx2
Id1
Slc2a1
Msx1
Wls
Sp5
Hand1
Ccnd2
Eomes
r = -0.09
pSom Correlation (r)
Gast Correlation (r)
Neuromesodermal Progenitor
Module
Gast
pSom
Both
Wnt3a
Grsf1
Wnt5b
T
Sp5
Rbp1
Id3
Spint1
Msx1
Laptm4b
Ddit4
Crabp1
Spint2
Id1
Msx2
r = 0.59
pSom Correlation (r)
Gast Correlation (r)
pmNC
mNC
CM
MC2
MC1